Satyam Kumar

Roll No.: 22CHB0B31
Chemical Engineering
Minor in ECE
National Institute Of Technology, Warangal

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Education

•National Institute of Technology, Warangal *2026*
Chemical Engineering B.Tech CGPA: 7.30

•Bapu Inter College, Gorakhpur *2020*
Intermediate, Uttar Pradesh Board 79.4%

Experience

• Oil and Natural Gas Corporation (ONGC), Eastern Offshore Asset *May-June 2024*
Industrial Internship-Production Processing of Oil & Natural Gas Facilities at Odalarevu Onshore Terminal (OOT) Kakinada

- Gained industrial exposure to **upstream, midstream, and downstream** operations in oil & gas production.
- **Explored major processing utilities** including Slug Catcher, Gas Compression, TEG-based Gas Dehydration, HC Dew Point Control, and MEG Reclamation Units.
- Optimized the Monoethylene Glycol (MEG) recovery process for efficient salt removal, reducing fouling, energy consumption, and maintenance costs while enhancing MEG recovery and reuse
- Interpreted and analyzed **PFDs, P&IDs, datasheets, and operating manuals**, applying core concepts of **thermodynamics, separation processes, and heat transfer** to real industrial systems.
- Learned about **hydrate prevention strategies, process safety (flare, HIPPS, fire water), and energy efficiency considerations** in gas terminals.

Personal Projects

•Synthesis of Nanofiber-Based Oxygen Carriers for Syngas Production via Chemical Looping Dry Reforming
Designed and synthesized Al_2O_3 nanofibers via electrospinning for application as oxygen carriers in CO_2 reforming of methane, achieving high CH_4 conversion, stable H_2/CO ratios, and improved coke resistance.

- **Tools & techniques used**: Electrospinning, Calcination, Fixed-bed CLR Reactor, SEM, XRD, BET, XPS
- Synthesized **Al_2O_3 nanofibers** from PVP–$Al(NO_3)_3$ and PAN–$Al(acac)_3$ precursors with **uniform Ni dispersion** ($\sim$ 3–4 nm), enabling $>90\%$ **CH_4** and $\approx 85\%$ **CO_2 conversion** with $H_2/CO \approx 1.0$–1.2.
- Demonstrated $\sim$ 40–50% **lower coke deposition** and enhanced catalyst stability, supporting **CO_2 utilization** and **clean hydrogen generation**.

Technical Skills and Interests

Languages: Python | MATLAB | SQL | C++
Simulation and Analysis Software: Aspen plus
Developer Tools: AutoCad | Ms Office | Visual Studio Code | PowerPoint | Canva
Soft Skills: Communication | Analytical thinking | Problem-solving | Team collaboration | Time management
Key Courses Studied: Heat Transfer | Mass Transfer | Chemical Engineering Thermodynamics | Fluid Mechanics |
Lab Skills: Mass Transfer lab | Heat Transfer lab | CRE lab, CP lab | FPM lab
Data Analysis & Visualization Tools: Pandas | NumPy | Matplotlib | Machine Learning | Seaborn | Power BI | MS Excel | Statistics | Data Warehousing & Analytics
Areas of Interest: Chemical Process Engineering | Process Optimization | Data Analysis | Advanced Process Control

Positions of Responsibility

• Undergraduate Representative- Athletics Society *Aug 2024 – April 2025*
–Led the planning and execution of **the Inter-NIT Athletics Meet (with over 1000 participants)**
–Managed logistics, scheduling, and coordination across multiple teams under tight timelines.
–Took **end-to-end ownership** of on-ground execution.

• Sponsorship Coordinator- SpringSpree, Cultural Fest (NITW's Flagship Event) *Oct 2024*
–Successfully pitched to the SBI branch manager, secured sponsorship, and enhanced brand visibility through high-traffic banner placements and dedicated stalls for credit card and loan promotions.

• Volunteer, Technozion, Annual Tech Fest *Aug 2024 – 2025*
–Coordinated 5+ technical & cultural events for 500+ students, ensuring smooth execution and management.
–Led promotional initiatives, boosting event attendance by 20% YoY

Achievements

•Awarded the Jindal Foundation Scholarship for academic excellence *2023*

•Selected for TES 47 SSB interview - 2021 *2021*